

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0635

Holland Cliff 138 kV
February 02, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0635. The Interconnection Customer proposes a 713.5 MW natural gas combined cycle facility interconnecting at the Holland Cliff 138 kV substation in the Maryland area. The request was received on June 22, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 252 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$85,367,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
WHITE PLAINS to HOLLAND CLIFF 138kV	CHALK POINT SUBSTATION to HOLLAND CLIFF 138kV line	300.7	233	100.0
WHITE PLAINS to SUNDERLAND 138kV L	HOLLAND CLIFF to CHALK PT 138kV line	299.2	230	100.0
UNIVERSITY OF NORTHERN IOWA to SHUTTLE 138kV	HOLLAND CLIFF to CHALK PT 138kV Line	295.9	216	100.0
HOLLAND CLIFF to CHALK PT SUBSTATION 138kV Line	HOLLAND CLIFF to WHITE PLAINS 138kV	295.8	237	100.0
SUNDERLAND to SHUTTLE 138kV	HOLLAND CLIFF to CHALK PT 138kV	293.7	225	100.0
BYRON BETHANY to CHALK PT 138kV line	CHALK PT 345/138 kV Auto	148.5	245	42.7
BYRON BETHANY to DUVALL 138kV	CHALK POINT 345/138 kV Transformer Bank	145.8	234	42.7
BUENA VISTA to MITCHELLVILLE 138kV	QUINCEY to PRIEST BRIDGE 138kV line	143.3	198	6.5
AQUEDUCT to DUVALL 138kV line	CHALK PT 345 kV/138 kV TX	133.7	218	42.7
AQUEDCT to OAK GROVE 138kV	CHALK POINT 345-138 kV Transformer Bank	131.1	208	42.7
BAY RIDGE to DIXIE VLY 138kV	HOLLAND CLIFF SUBSTATION to CHALK PT 138kV line	118.2	451	40.1
DIXIE VALLEY to STRANGER CRK 138kV	CHALK POINT to HOLLAND CLIFF 138kV line	117.1	467	40.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
STRANGER CREEK to UNIVERSITY OF NORTHERN IOWA 138kV line	HOLLAND CLIFF to CHALK PT 138kV	116.1	484	40.1
TAZEWELL to PORT ROBSON 138kV	DIXLETA SUBSTATION to TABER SUBSTATION 345kV L	105.8	521	5.3
RUSH to COLESVILLE 138kV	GLENANNAN to TABER 345kV Line	104.2	503	7.0
TAZEWELL to TABER SUBSTATION 138kV	TABER to DIXLETA 345kV Line	103.3	603	5.3
MITCHELLVILLE to QUINCEY 138kV Line	PRIEST BRIDGE to QUINCEY 138kV	103.0	289	6.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WHITE PLAINS to HOLLAND CLIFF 138kV	Rebuild line with higher-capacity conductor	\$3,509,000	100% (\$3,509,000)
WHITE PLAINS to SUNDERLAND 138kV L	Rebuild line with higher-capacity conductor	\$2,602,000	100% (\$2,602,000)
UNIVERSITY OF NORTHERN IOWA to SHUTTLE 138kV	Rebuild line with higher-capacity conductor	\$2,732,000	100% (\$2,732,000)
CHALK PT to HOLLAND CLIFF 138kV Line	Rebuild line with higher-capacity conductor	\$7,073,000	100% (\$7,073,000)
SUNDERLAND to SHUTTLE 138kV	Rebuild line with higher-capacity conductor	\$23,890,000	100% (\$23,890,000)
BYRON BETHANY to CHALK PT 138kV line	Rebuild line with higher-capacity conductor	\$15,889,000	87% (\$13,789,000)
BYRON BETHANY to DUVALL 138kV	Rebuild line with higher-capacity conductor	\$17,028,000	87% (\$14,777,000)
BUENA VISTA to MITCHELLVILLE 138kV	Rebuild line with higher-capacity conductor	\$8,417,000	29% (\$2,429,000)
AQUEDUCT to DUVALL 138kV line	Rebuild line with higher-capacity conductor	\$7,310,000	87% (\$6,343,000)
OAK GROVE to AQUEDUCT 138kV	Rebuild line with higher-capacity conductor	\$1,945,000	87% (\$1,688,000)
BAY RIDGE to DIXIE VLY 138kV	Rebuild line with higher-capacity conductor	\$3,282,000	65% (\$2,136,000)
STRANGER CRK to DIXIE VLY 138kV	Rebuild line with higher-capacity conductor	\$1,828,000	65% (\$1,190,000)
UNIVERSITY OF NORTHERN IOWA to STRANGER CRK 138kV	Rebuild line with higher-capacity conductor	\$2,910,000	65% (\$1,893,000)
TAZEWELL to PORT ROBSON 138kV	Reconductor line	\$758,000	25% (\$186,000)
RUSH to COLESVILLE 138kV	Reconductor line	\$3,626,000	15% (\$558,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
TAZEWELL to TABER 138kV	Reconductor line	\$1,274,000	25% (\$313,000)
QUINCEY to MITCHELLVILLE 138kV L	Reconductor line	\$897,000	29% (\$259,000)
	Total allocated to project		\$85,367,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.